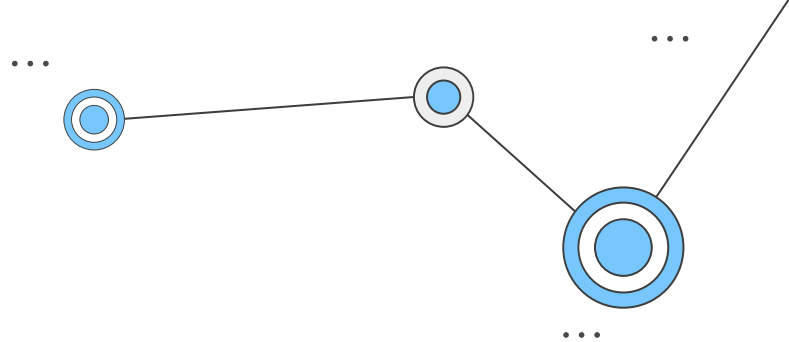
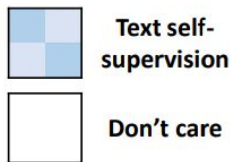
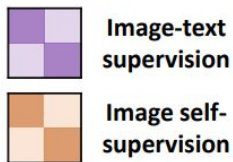
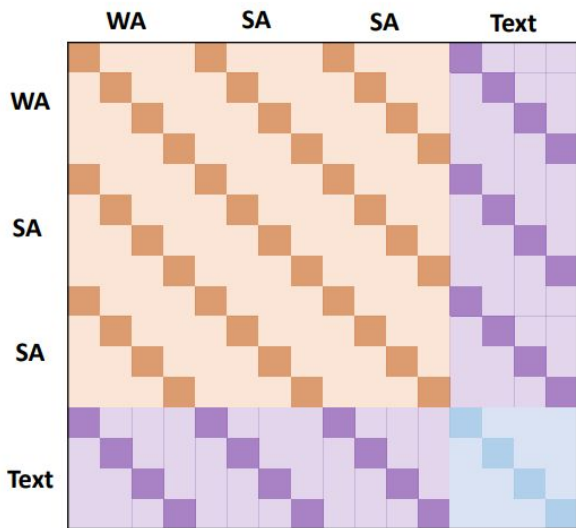




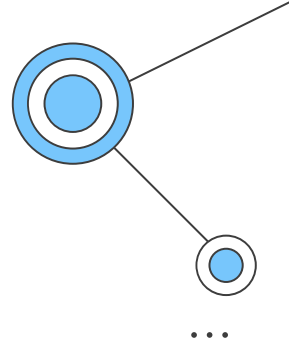
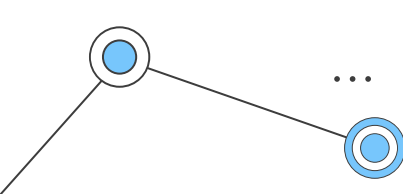
UniCLIP: Unified Framework for Contrastive Language-Image Pre-training (NeurIPS, 2022)



이미지처리팀

류채은*, 강인하, 김병현, 김준철, 안종식, 최승준,

허정원



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Notations**

02

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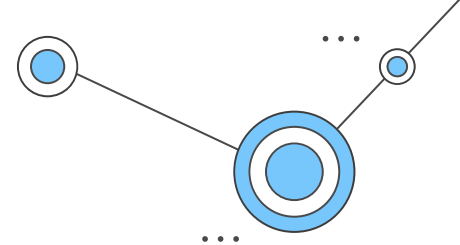
03

**Experiments/
Ablations Study**

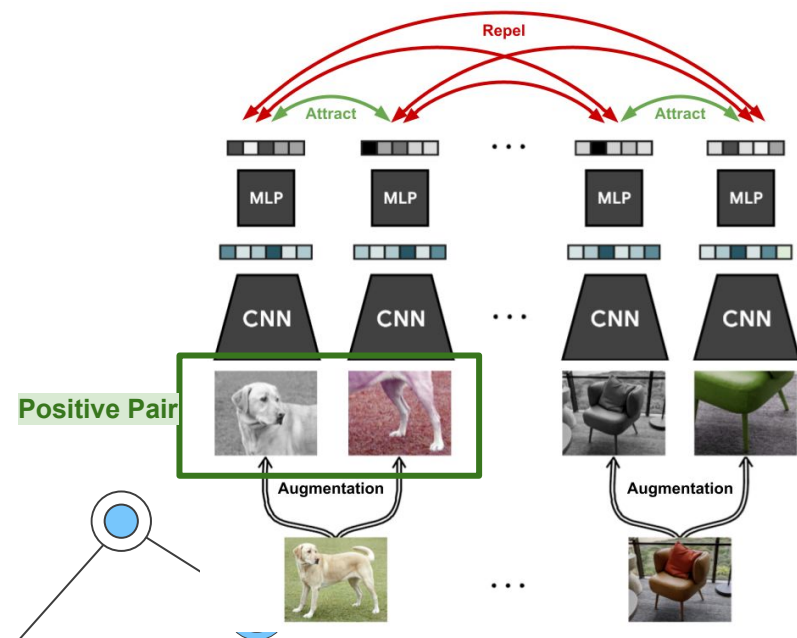
04

Conclusion

Background/Notations

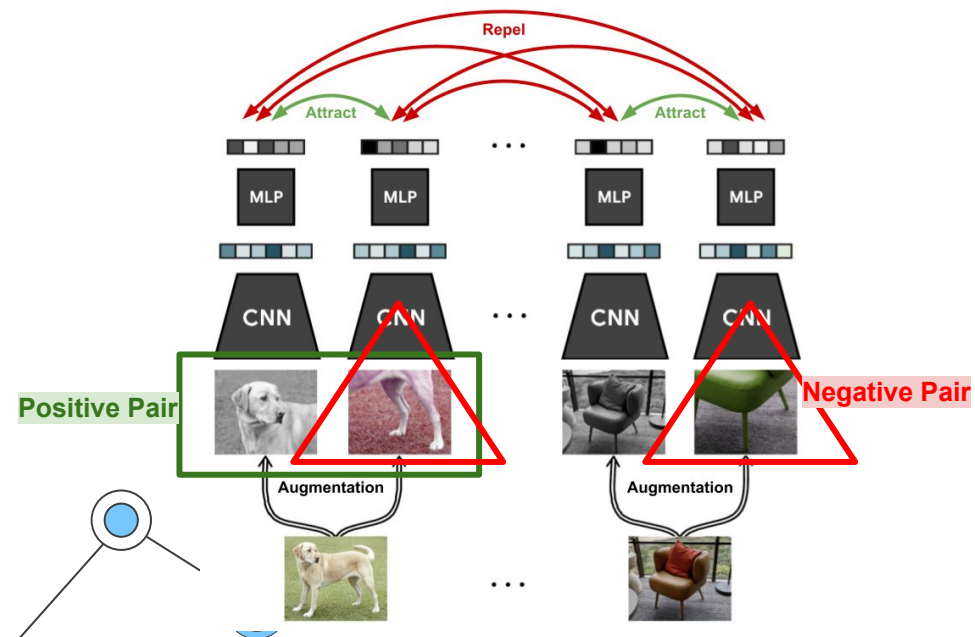


Contrastive Learning (SimCLR)



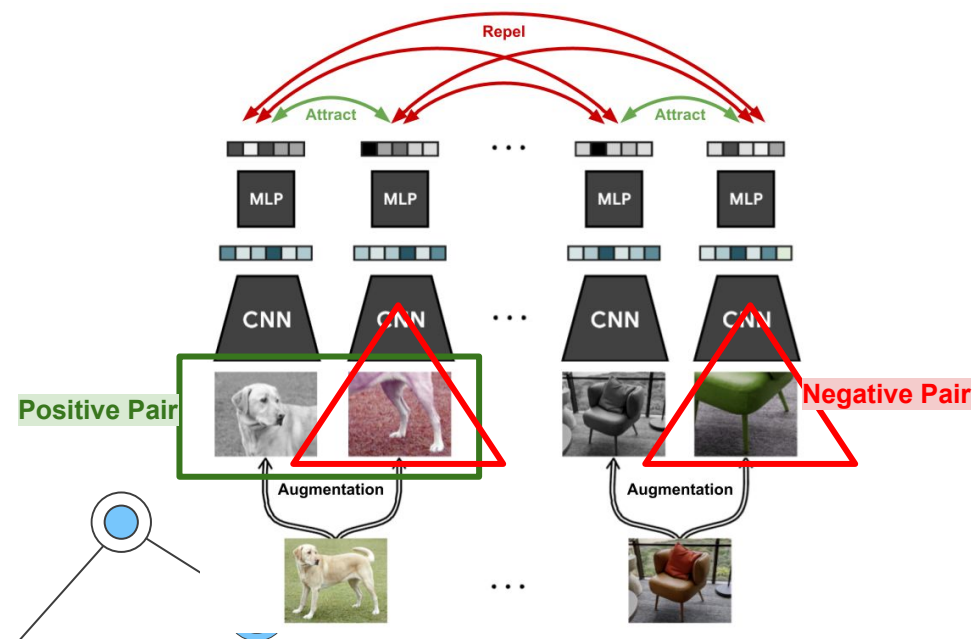
Background/Notations

Contrastive Learning (SimCLR)

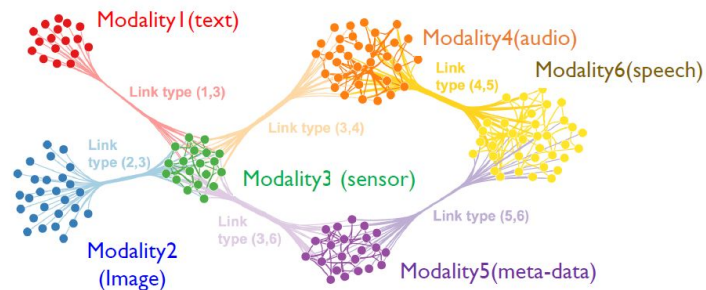


Background/Notations

Contrastive Learning (SimCLR)



Domains



Intra-domain



text & text

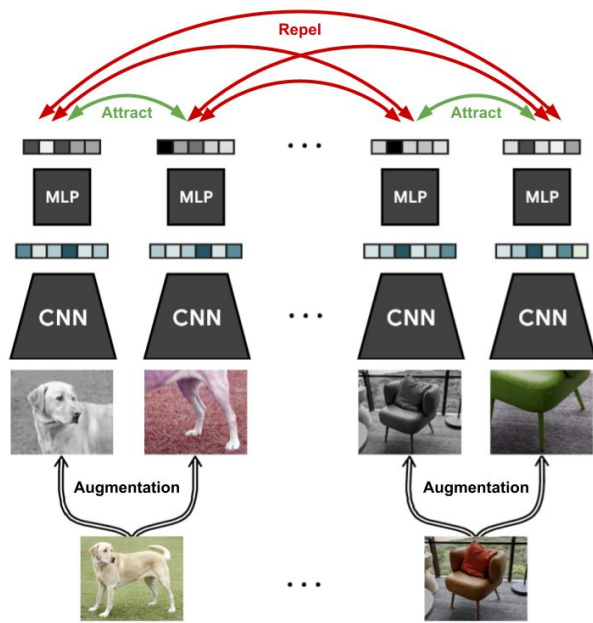
Inter-domain



& "Flowers of *yellow* color."

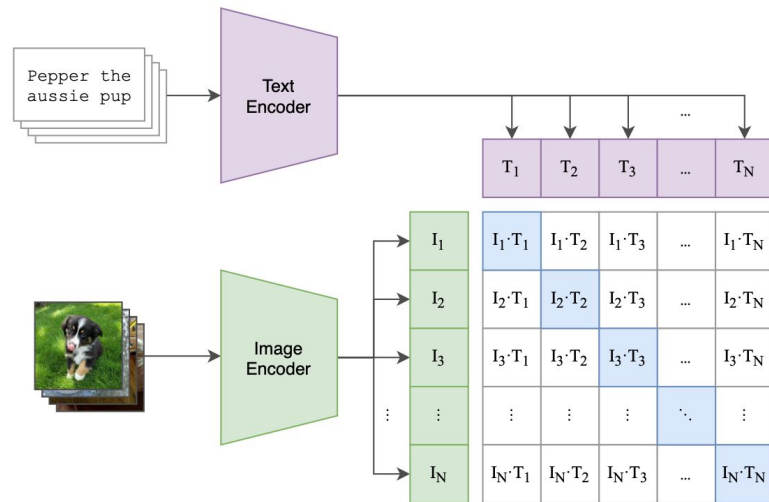
Background/Notations

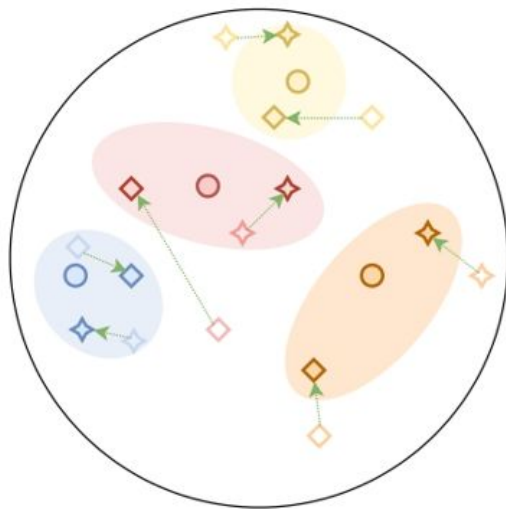
Contrastive Learning (SimCLR)



Multimodal Embedding (CLIP)

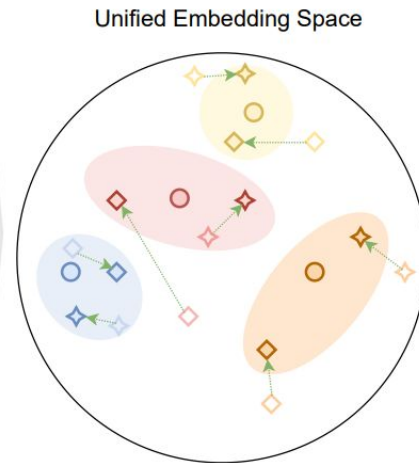
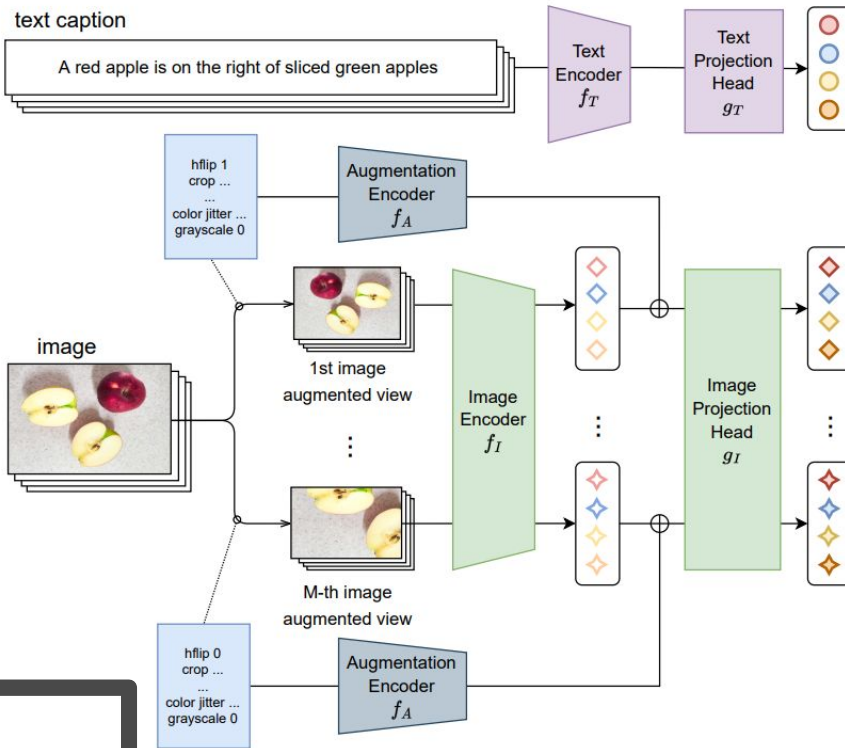
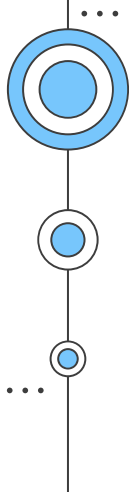
(1) Contrastive pre-training





Goal of the paper

Build a contrastive image-text pre-training framework where the contrastive learning of all possible intra-domain and inter-domain pairs is defined in the same single unified embedding space



⊕ concat

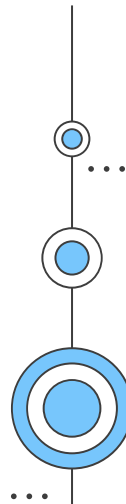
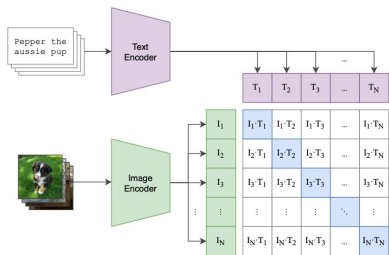
→ misalignment adjustments by UniCLIP

Framework Overview

Build a contrastive image-text pre-training framework where the contrastive learning of all possible intra-domain and inter-domain pairs is defined in the same single unified embedding space

Open AI, CLIP

(1) Contrastive pre-training



Contributions

... 01

**Augmentation-aware
Feature Embedding**

02

MP-NCE Loss

03

Goal of the paper
**Domain dependent
similarity measure**

Build a contrastive image-text pre-training framework where the contrastive learning of all possible intra-domain and inter-domain pairs is defined in the same single unified embedding space

Contributions

01

**Augmentation-aware
Feature Embedding**

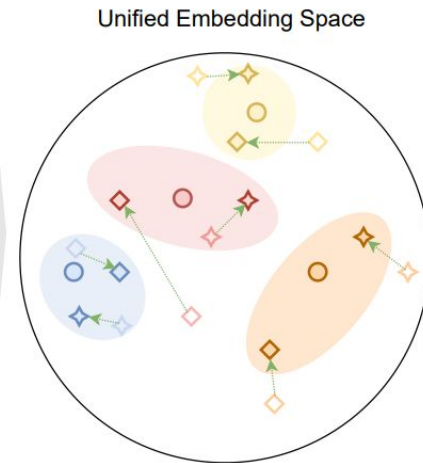
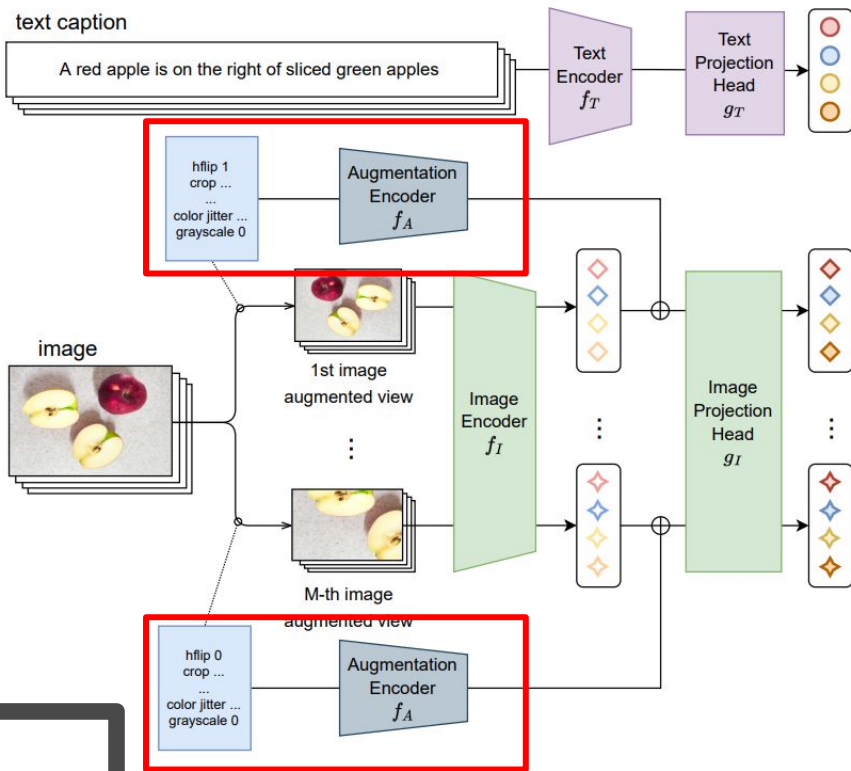
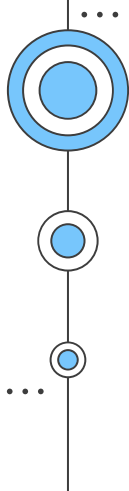
02

MP-NCE Loss

03

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Build a contrastive image-text pre-training framework where the contrastive learning of all possible intra-domain and inter-domain pairs is defined in the same single unified embedding space



⊕ concat

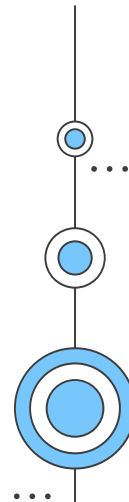
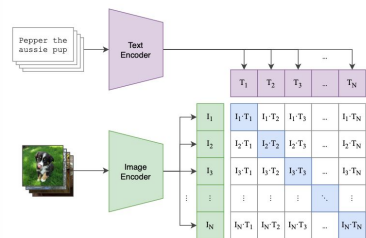
→ misalignment adjustments by UniCLIP

Framework Overview

Build a contrastive image-text pre-training framework where the contrastive learning of all possible intra-domain and inter-domain pairs is defined in the same single unified embedding space

Open AI, CLIP

(1) Contrastive pre-training

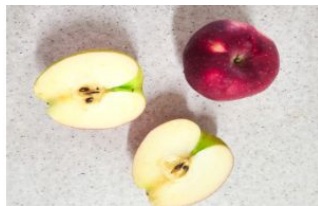


01. Augmentation-aware feature embedding

[Motivation]

Challenge:

Misalignments can occur between the image-text semantics when applying image augmentations



A red apple is on the right of sliced green apples

Horizontal Flip



[Augmented Image] ...

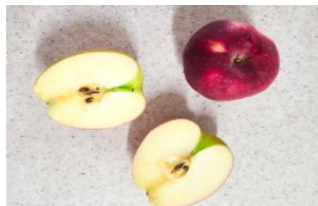
*A red apple is on the **right** of sliced green apples*

01. Augmentation-aware feature embedding

[Motivation]

Challenge:

Misalignments can occur between the image-text semantics when applying image augmentations



*A red apple is on the right
of sliced green apples*

Grayscale

[Augmented Image]



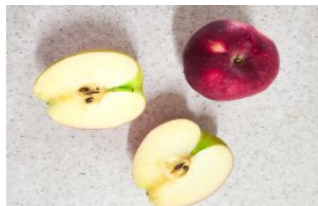
*A ~~red~~ apple is on the right
of sliced ~~green~~ apples*

01. Augmentation-aware feature embedding

[Motivation]

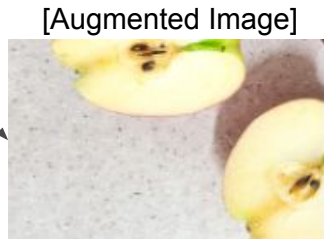
Challenge:

Misalignments can occur between the image-text semantics when applying image augmentations



A red apple is on the right of sliced green apples

Crop



[Augmented Image]

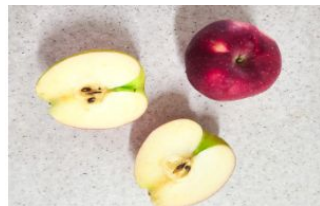
~~A red apple~~ is on the right of sliced green apples

01. Augmentation-aware feature embedding

[Motivation]

Challenge:

Misalignments can occur between the image-text semantics when applying image augmentations

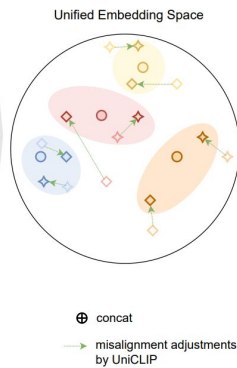
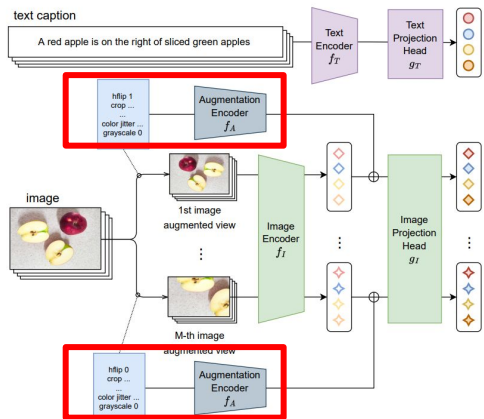


A red apple is on the right of sliced green apples



Let's make the model be aware of misalignments caused by augmentation!

01. Augmentation-aware feature embedding

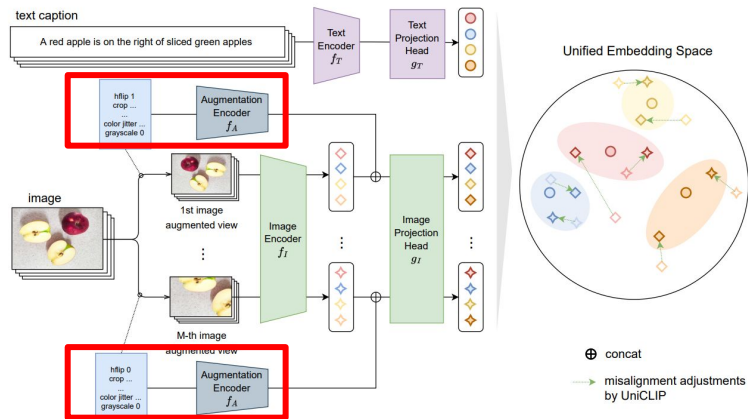


Describe a 11 dimensional real vector containing information about how much each basic transformation is applied to data.

*Method 1) **RandomResizedCrop**: (x,y,w,h)*

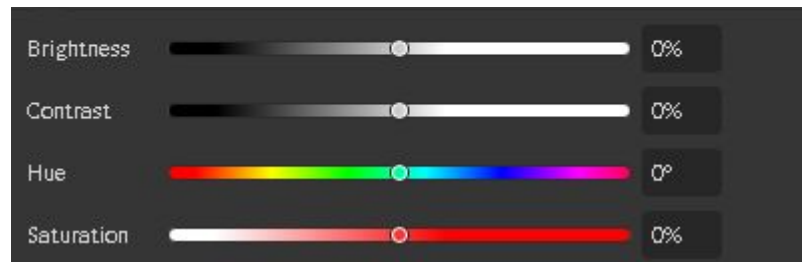


01. Augmentation-aware feature embedding

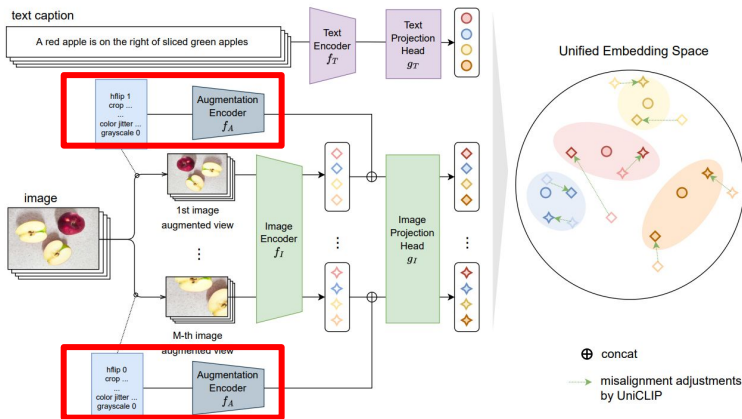


Describe a 11 dimensional real vector containing information about how much each basic transformation is applied to data.

*Method 2) **Color Jitter:** (brightness, contrast, saturation, hue)*



01. Augmentation-aware feature embedding

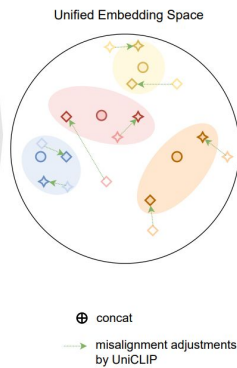
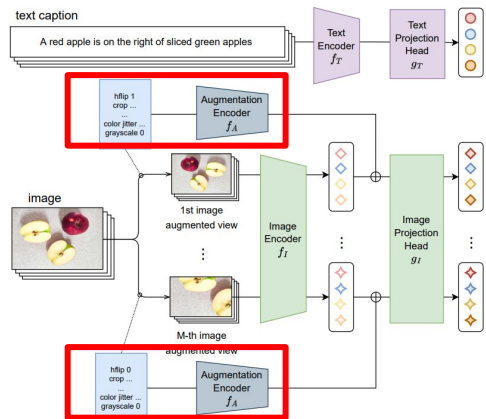


Describe a 11 dimensional real vector containing information about how much each basic transformation is applied to data.

*Method 3) **Gaussian Blur: (StDev)***



01. Augmentation-aware feature embedding



Describe a 11 dimensional real vector containing information about how much each basic transformation is applied to data.

Method 4) Horizontal Flip: (0 or 1)

Method 5) Grayscale Convert: (0 or 1)



Horizontal Flip: [0]



Horizontal Flip: [1]

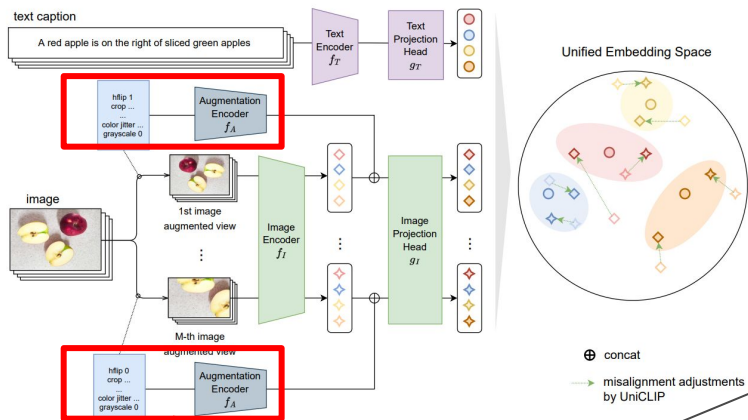


Grayscale Convert: [0]



Grayscale Convert: [1]

01. Augmentation-aware feature embedding



Describe a 11 dimensional real vector containing information about how much each basic transformation is applied to data.

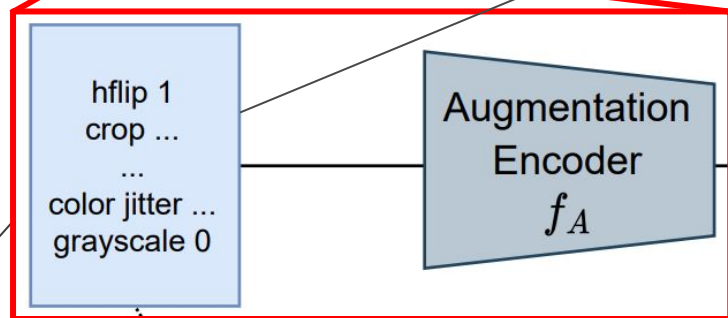
*Method 1) **RandomResizedCrop**: (x,y,w,h)*

*Method 2) **Color Jitter**: (brightness, contrast, saturation, hue)*

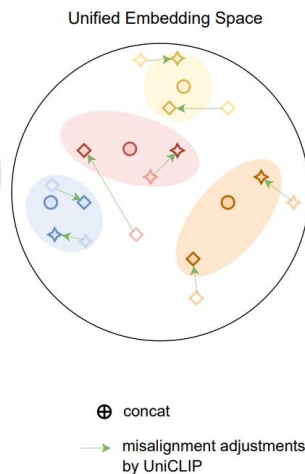
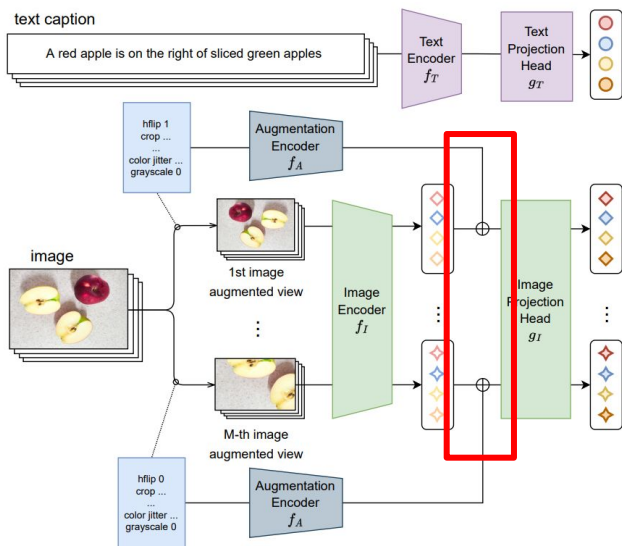
*Method 3) **Gaussian Blur**: (StDev)*

*Method 4) **Horizontal Flip**: (0 or 1)*

*Method 5) **Grayscale Convert**: (0 or 1)*



01. Augmentation-aware feature embedding



Describe a 11 dimensional real vector containing information about how much each basic transformation is applied to data.

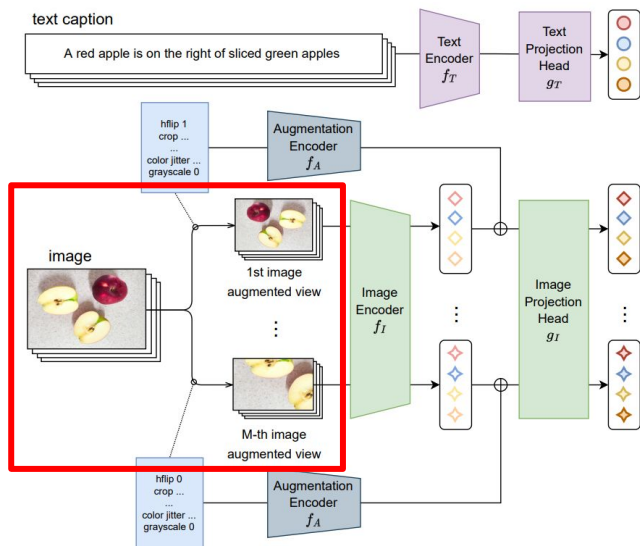
- Method 1) **RandomResizedCrop**: (x,y,w,h)
- Method 2) **Color Jitter**: (brightness, contrast, saturation, hue)
- Method 3) **Gaussian Blur**: (StDev)
- Method 4) **Horizontal Flip**: (0 or 1)
- Method 5) **Grayscale Convert**: (0 or 1)

f_I : Augmentation-aware

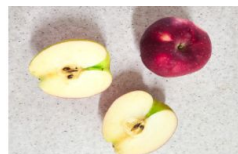
g_I : Augmentation-agnostic

In this way, the encoder can fully enjoy the benefits of data augmentation and generalizes better, while the projection head is still able to correct inter-domain misalignments caused by the augmentations.

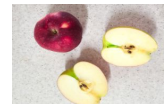
01. Augmentation-aware feature embedding



image



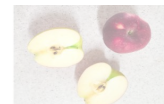
Strongly Augmented Image Embedding 1



Strongly Augmented Image Embedding 2



Weakly Augmented Image Embedding

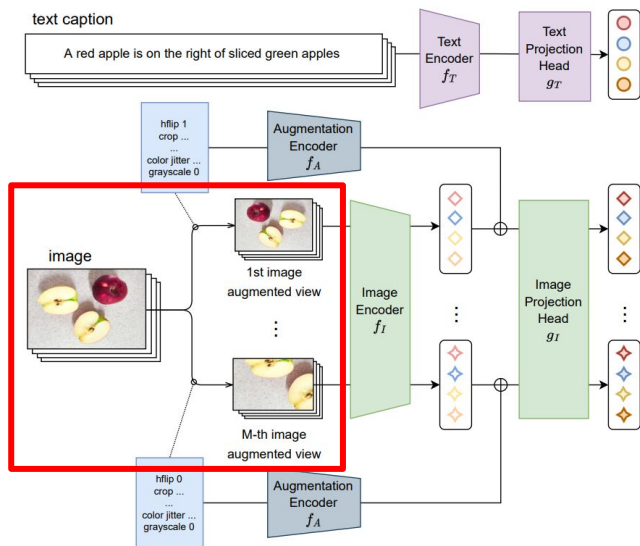


text

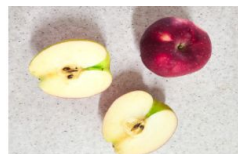
A red apple is on
the right
of sliced green
apples

Text Embedding

01. Augmentation-aware feature embedding



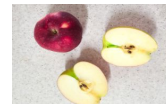
image



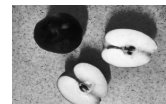
Strongly Augmented Image Embedding 1

Strongly Augmented Image Embedding 2

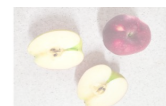
Weakly Augmented Image Embedding



SA



SA

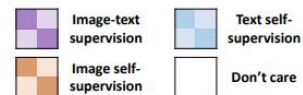
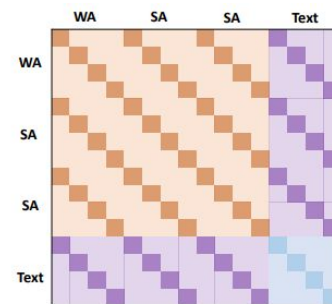


WA

text

A red apple is on the right of sliced green apples

Text Embedding **Text**



Contributions

... 01

Augmentation-aware
Feature Embedding

02

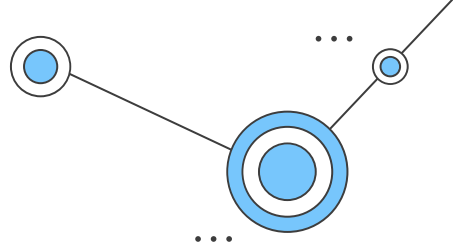
MP-NCE Loss

03

Goal of the paper
Domain dependent
similarity measure

Build a contrastive image-text pre-training framework where the contrastive learning of all possible intra-domain and inter-domain pairs is defined in the same single unified embedding space

02. Multi Positive NCE Loss



$$\mathcal{L}_i^{\text{MP-NCE}} = \mathbb{E}_{p \in P_i \cup \{i\}} \left[-w_{\mathcal{D}(i,p)} \log \frac{s_{i,p}}{s_{i,p} + \sum_{n \in N_i} s_{i,n}} \right]$$

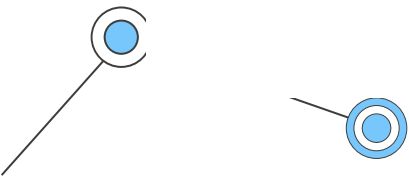
[Notations]

For an i -th embedding z_i in a batch of embeddings $\{z_i\}_i$, let P_i be the set of all positive sample indices of the i -th sample excluding i itself and N_i be the set of all negative sample indices of the i -th sample.

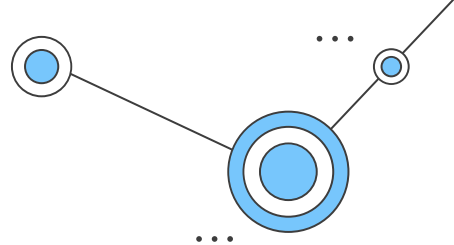
$$P_i = \{j | (z_i, z_j) \text{ is a positive pair and } j \neq i\} \quad (1)$$

$$N_i = \{j | (z_i, z_j) \text{ is a negative pair}\} \quad (2)$$

$$s_{i,j} = \text{sim}(z_i, z_j)$$



02. Multi Positive NCE Loss

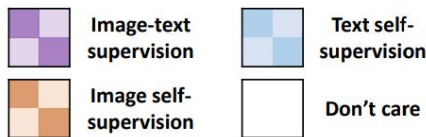
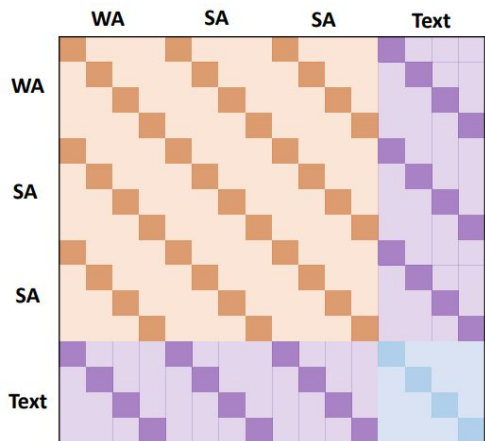


$$\mathcal{L}_i^{\text{MP-NCE}} = \mathbb{E}_{p \in P_i \cup \{i\}} \left[-w_{\mathcal{D}(i,p)} \log \frac{s_{i,p}}{s_{i,p} + \sum_{n \in N_i} s_{i,n}} \right]$$

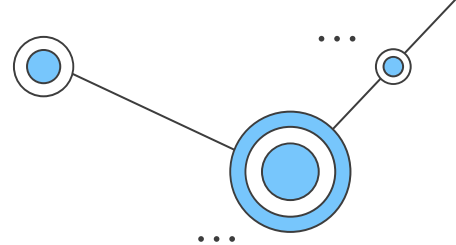
$w_{\mathcal{D}(i,p)}$: domain-specific balancing hyperparameter which makes each inter-domain and intra-domain supervision equally contributes to the loss.

$w_D(i, p)$ {

- 1/9, If (z_i, z_p) : Image-image pair
- 1/6, If (z_i, z_p) : Image-text pair
- 1/1, If (z_i, z_p) : text-text pair



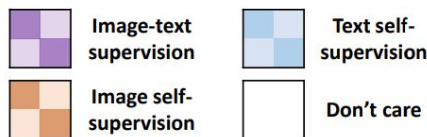
02. Multi Positive NCE Loss



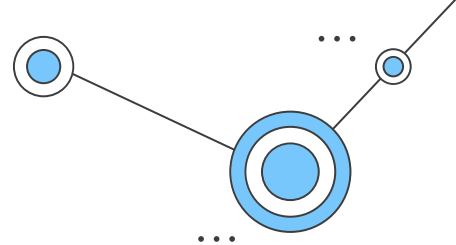
$$\mathcal{L}_i^{\text{MP-NCE}} = \mathbb{E}_{p \in P_i \cup \{i\}} \left[-w_{\mathcal{D}(i,p)} \log \frac{s_{i,p}}{s_{i,p} + \sum_{n \in N_i} s_{i,n}} \right]$$

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 1/6, If (z_i, z_p) : Image-text pair
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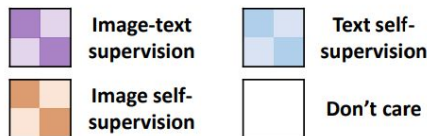
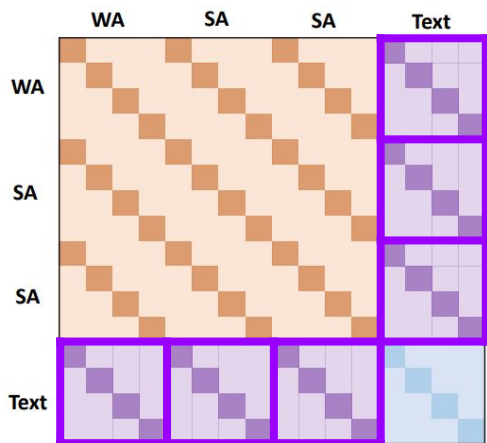
02. Multi Positive NCE Loss



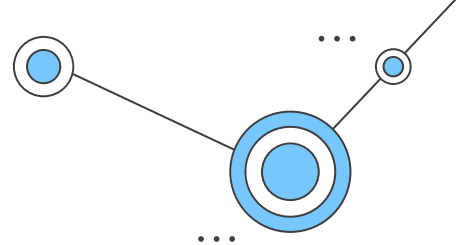
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 1/1, If (z_i, z_p) : text-text pair

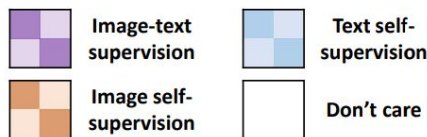
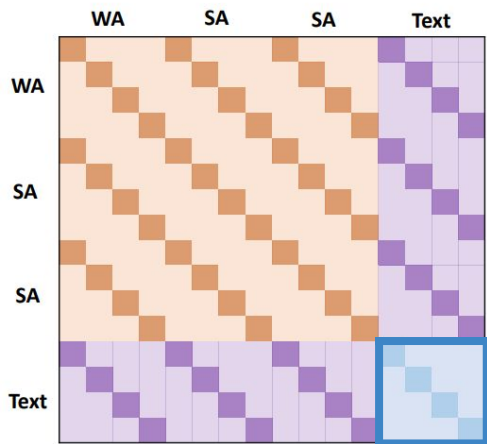


02. Multi Positive NCE Loss



$$\mathcal{L}_i^{\text{MP-NCE}} = \mathbb{E}_{p \in P_i \cup \{i\}} \left[-w_{\mathcal{D}(i,p)} \log \frac{s_{i,p}}{s_{i,p} + \sum_{n \in N_i} s_{i,n}} \right]$$

$w_{\mathcal{D}(i,p)}$: domain-specific balancing hyperparameter which makes each inter-domain and intra-domain supervision equally contributes to the loss.



- 1/9, If (z_i, z_p) : Image-image pair
- 1/6, If (z_i, z_p) : Image-text pair
- 1/1, If (z_i, z_p) : text-text pair

Contributions

... 01

Augmentation-aware
Feature Embedding

02

MP-NCE Loss

03

Goal of the paper
Domain dependent
similarity measure

Build a contrastive image-text pre-training framework where the contrastive learning of all possible intra-domain and inter-domain pairs is defined in the same single unified embedding space

		
"motorcycle front wheel"	"thumbnail for version as of 21 57 29 june 2010"	"file frankfurt airport skyline 2017 05 jpg"
		
"file london barge race 2 jpg"	"wild boar head portrait forest creature boar"	"st oswalds way and shops"

Domain dependent similarity measure

[Cosine Similarity]

$$\text{Similarity}(p, q) = \cos \theta = \frac{p \cdot q}{\|p\| \|q\|} = \frac{\sum_{i=1}^n p_i q_i}{\sqrt{\sum_{i=1}^n p_i^2} \sqrt{\sum_{i=1}^n q_i^2}}$$

[Proposed domain-dependent similarity score]

$$s_{i,j} = \exp \left(\frac{1}{\tau_{\mathcal{D}(i,j)}} \left(\frac{z_i^\top z_j}{\|z_i\| \|z_j\|} - b_{\mathcal{D}(i,j)} \right) \right)$$

Temperature:
extending the range of
similarity score

Offset:
classify the pair i, j as
positive if the cosine
similarity between the pair
is greater than b , and
negative otherwise

Expect that the optimal threshold will be learned
by the model

3 different temperatures
3 different offsets learned!

3: (image-image, image-text, text-text)

The threshold can be different depending on whether the sampled data pair is an intra-domain pair or an inter-domain pair. (It would be easier to classify intra-domain positive pairs than inter-domain positive pairs)

Experiments

Table 1: Zero-shot image classification performance and linear probing performance on 11 downstream datasets. [†]Results reported in the original paper.

Method	Pre-train dataset	Pets	CIFAR-10	CIFAR-100	SUN397	Food-101	Flowers	Cars	Caltech-101	Aircraft	DTD	ImageNet	Average
<i>Zero-shot classification:</i>													
CLIP-ViT-B/32	YFCC15M	19.4	62.3	33.6	40.2	33.7	6.3	2.1	55.4	1.4	16.9	31.3	27.5
SLIP-ViT-B/32	YFCC15M	28.3	72.2	45.3	45.1	44.7	6.8	2.9	65.9	1.9	21.8	38.3	33.9
DeCLIP-ViT-B/32	YFCC15M	30.2	72.1	39.7	51.6	46.9	7.1	3.9	70.1	2.5	24.2	41.2	35.4
UniCLIP-ViT-B/32	YFCC15M	32.5	78.6	47.2	50.4	48.7	8.1	3.4	73.0	2.8	23.3	42.8	37.3
<i>Linear probing:</i>													
CLIP-ViT-B/32	YFCC15M	71.2	89.2	72.1	70.1	71.4	93.2	34.9	84.3	29.7	60.9	61.1	67.1
SLIP-ViT-B/32	YFCC15M	75.4	90.5	75.3	73.5	77.1	96.1	43.0	87.2	34.1	71.1	68.1	71.9
DeCLIP-ViT-B/32	YFCC15M	76.5	88.6	71.6	75.9	79.3	96.7	42.6	88.0	32.6	69.1	69.2	71.8
UniCLIP-ViT-B/32	YFCC15M	83.1	92.5	78.2	77.0	81.3	97.1	49.8	88.9	36.2	72.8	70.8	75.2

Table 2: ImageNet-1k fine-tuning accuracy for the models pre-trained on YFCC15M.

Method	Accuracy
CLIP-ViT-B/32	72.27
SLIP-ViT-B/32	75.64
DeCLIP-ViT-B/32	74.34
UniCLIP-ViT-B/32	76.54

Ablation Studies

(a) **Image projection head types.** One weak and two strong image augmentations are used.

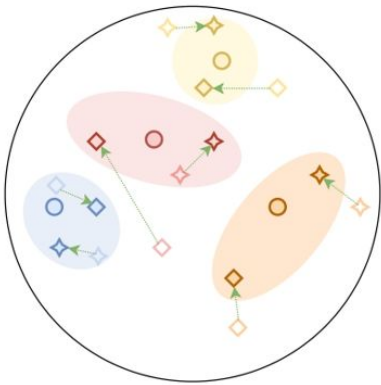
Augmentation embedding	Head type	Accuracy
✗	MLP 3 layers	24.01
	MLP 6 layers	23.62
	1 ResBlock	24.76
	3 ResBlocks	24.46
✓	Linear layer	24.68
	MLP 3 layers	24.54
	MLP 6 layers	24.15
	1 ResBlock	27.67
	3 ResBlocks	27.84

Making the image projection head augmentation-aware improves the performance for all head types as the head can handle the inter-domain misalignments.

(b) **Augmentation configurations.** 1-ResBlock head is used for no augmentation embedding config and 3-ResBlock head is used with augmentation embedding.

Augmentation embedding	Augmentation	Accuracy
✗	3 weak	24.49
	1 weak, 2 strong	24.76
	3 strong	22.60
✓	3 weak	23.40
	1 weak, 2 strong	27.84
	3 strong	26.43

Using only strongly augmented images severely degrades the performance without augmentation embedding, because strong augmentations will generate more image-text misalignments.



Conclusion

- ❖ We have proposed **UniCLIP, a unified framework** for visual-language pre-training by integrating contrastive losses defined across multiple domains into a single universal space.
- ❖ Although we have experimented with **vision-language multimodal datasets** only, the proposed UniCLIP framework can be easily extended to other types of multimodal datasets because it is designed in a modality-agnostic way except for the augmentation encoding part.
- ❖ We leave it for **future work** to see how well the UniCLIP framework works with **various types of multimodal datasets**.